

AI-AGENT-DRIVEN MIGRATION FRAMEWORK

Synapse → Databricks

5 specialized agents, 1 orchestrator, and two human approval gates

5

AGENTS

2

HUMAN GATES

3

MCP SERVERS

How the migration flows

Orchestrator sequences five agents between two human approval gates



The five agents

Each answers one question in the migration

1	Discovery <i>What exists in Synapse?</i>	Read-only inventory of pipelines, data flows, sources, sinks and linked services.
2	Analysis <i>What does it do?</i>	Maps join keys, join types, aggregations and business rules — flags risks like row loss.
3	Design <i>How should it look in Databricks?</i>	Translates the Synapse pipeline into a Delta table + PySpark target design.
4	Implementation <i>Build it.</i>	First agent with write access — generates notebooks, workflows, SQL and tests.
5	Validation <i>Did we build it right?</i>	Compares schema, row counts, nulls and aggregations against the Synapse source.

Example: One Aggregate, Start to Finish

How the agent turns a Synapse activity into working framework configuration.

`read_pipeline("banking_pipeline") → JSON`

```
{
  "activities": [{
    "name": "Aggregate Transactions",
    "type": "Aggregate",
    "groupBy": ["customer_id"],
    "aggregations": {
      "transaction_count":
        "count(transaction_id)",
      "total_transaction_amount":
        "sum(amount)"
    }
  }]
}
```



*Agent reasons:
"This Aggregate
maps to
build_aggregation
()"*

`generate_config() → GOLD_CONFIG`

```
"fact_transactions": {
  "source": "transactions",
  "group_by": ["customer_id"],
  "aggregations": {
    "transaction_count": {
      "function": "count",
      "column": "transaction_id"
    },
    "total_transaction_amount": {
      "function": "sum",
      "column": "amount"
    }
  }
}
```

Validation & the Migration Report

Every migration is checked against the source before it's called done — and the result is written down.

Validation Agent — Output

dim_customer

- ✔ Record count
- ✔ Schema
- ✔ Primary key
- ✔ Null checks

dim_account

- ✔ Record count
- ✔ Schema
- ✔ Primary key

fact_transactions

- ✔ Aggregation
- ✔ Transaction count
- ✔ Total amount

Overall: PASS



migration_report.md

Source	Azure Synapse
Target	Databricks
Pipeline	Banking Pipeline
Gold objects	3 dims + 1 fact
Validation	PASS
Warning	transaction_status = UNKNOWN
Human approval	Approved

Permission model

Write access is earned — only Implementation can touch Databricks or Git

Agent	Synapse	Databricks	Git
Discovery	Read	—	Read
Analysis	Read	Read	Read
Design	Read	Read	Read
Implementation	Read	Write	Write
Validation	Read	Run / Read	Read
Orchestrator	Controlled	Controlled	Controlled

Deployment always stays behind Human Review 2 — no agent can deploy to production.

MCP servers connect agents to tools

Three scoped servers, each with its own tool set

Synapse MCP

READ-ONLY

- `list_pipelines()`
- `get_pipeline()`
- `get_dataflow()`
- `get_schema()`

Databricks MCP

SCOPED BY AGENT

- `list_tables()`
- `create_notebook()`
- `run_workflow()`
- `get_job_status()`

Git MCP

SCOPED BY AGENT

- `read_repository()`
- `create_branch()`
- `write_file()`
- `create_pull_request()`

How It Fits The Pipeline

It maps directly onto transaction_pipeline

- Discovers real sources: customers, accounts, branches, transactions
- Analyzes the exact, conditions and joins that were silently dropping rows
- Designs the Delta + PySpark target before any code is written
- Validates row counts, nulls and sums against Synapse before sign-off

Agents → MCP → shared context → human-in-the-loop → migration.